Stage 8 Maths — Practice Test 3 — Mark Scheme

Paper 1: 60 minutes · no calculator · Paper 2: 60 minutes · calculator allowed · Total 100 marks

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

Paper 1 — 60 minutes — 50 marks

1 **[1]**

(a) Work out $5 + 3 \times 2^2$

Answer: 17

Notes / working: Indices first: $2^2 = 4$. Then multiply: $3 \times 4 = 12$. Then add: $5 + 12 =$ **17**. Common wrong answer: $8 \times 4 = 32$ (adding first).

(b) Work out $(-6) \times (-4) \div (-3)$ **[1]**

Answer: -8

Notes / working: $(-6) \times (-4) = +24$ (two negatives make a positive). $24 \div (-3) =$ **−8** (positive ÷ negative is negative).

2 **[1]**

Find the value of n.

$7^5 \times 7^3 \div 7^8 = 7^n$

Answer: 0

Notes / working: Multiply → add the indices: $5 + 3 = 8$. Divide → subtract: $8 - 8 = 0$. So **n = 0** (and $7^0 = 1$). Do not write n = 1: the value of the expression is 1, but the index is 0.

3 **[1]**

(a) Write 84 as a product of its prime factors.

Answer: 1 mark for $2 \times 2 \times 3 \times 7$ or $2^2 \times 3 \times 7$ (factors in any order). Do not accept answers that include 1 or a non-prime factor such as $4 \times 3 \times 7$.

Notes / working: $84 = 2 \times 42 = 2 \times 2 \times 21 = 2 \times 2 \times 3 \times 7$, so **$84 = 2^2 \times 3 \times 7$**. A factor tree or repeated division by primes both work. Check: $4 \times 3 \times 7 = 84$.

(b) At a railway station, a train to Riverside leaves every 12 minutes and a train to Hilltop leaves every 18 minutes. **[2]**

Two trains, one to each town, leave together at 09:00.

At what time will a train to Riverside and a train to Hilltop next leave together? Show your working.

Answer: 09:36 or 9:36 or 0936 or 09.36 or 9.36 or 09:36 am or 9:36 am or 9:36am or 09:36am
Notes / working: We need the lowest common multiple of 12 and 18.
Multiples of 12: 12, 24, **36**, 48 … Multiples of 18: 18, **36**, 54 …
(or $12 = 2^2 \times 3$ and $18 = 2 \times 3^2$, so LCM = $2^2 \times 3^2 = 36$.)
09:00 + 36 minutes = **09:36**.
Marking: 1 mark for 36 seen (LCM); 1 mark for 09:36. Common mistake: using the HCF (6) and answering 09:06.

---

4 **[1]**

Which of these numbers is an integer but **not** a natural number?

- ☐ A) 7
- ☐ B) −3
- ☐ C) 0.5
- ☐ D) 2/3

Answer: B) −3
Notes / working: Natural numbers are the counting numbers 1, 2, 3, … Integers are whole numbers, including negative ones. 7 is natural (and an integer); 0.5 and 2/3 are rational but not integers. **−3** is an integer but not a natural number.

---

5 **[1]**

(a) Work out $\sqrt[3]{(-64)}$

Answer: -4
Notes / working: $(-4) \times (-4) \times (-4) = 16 \times (-4) = -64$, so $\sqrt[3]{(-64)} = \mathbf{-4}$. Negative numbers DO have a cube root (but not a square root).

(b) Youssef says, '−49 has two square roots: 7 and −7.' **[1]**

Is he correct? Explain your answer.

Answer: 1 mark for 'No' with a correct reason, e.g. $7^2 = 49$ and $(-7)^2 = 49$, so 7 and −7 are the square roots of 49, not −49; OR a square of any number cannot be negative. Do not accept 'No' with no reason.
Notes / working: He is **not correct**. 7 × 7 = 49 and (−7) × (−7) = 49. Squaring a positive or a negative number always gives a positive answer, so no number squared gives −49. The two square roots he wrote (7 and −7) belong to **+49**.

6 **[1]**

(a) Work out 46 × 0.01

> **Answer:** 0.46
> **Notes / working:** Multiplying by 0.01 is the same as dividing by 100: 46 ÷ 100 = **0.46**.

(b) Work out 0.75 ÷ 0.01 **[1]**

> **Answer:** 75
> **Notes / working:** Dividing by 0.01 is the same as multiplying by 100: 0.75 × 100 = **75**. Check: 75 × 0.01 = 0.75 ✓. Dividing by a number less than 1 makes the answer bigger.

---

7 **[2]**

(a) Work out 4 1/3 − 1 3/4
Give your answer as a mixed number in its simplest form. Show your working.

> **Answer:** 2 7/12
> **Notes / working:** Estimate: about 4 − 2 = 2.
> Method 1 (improper fractions): 4 1/3 = 13/3 = 52/12 and 1 3/4 = 7/4 = 21/12. 52/12 − 21/12 = 31/12 = **2 7/12**.
> Method 2: 4 1/3 − 1 = 3 1/3 = 3 4/12; 3 4/12 − 9/12 = 2 16/12 − 9/12 = 2 7/12.
> Marking: 1 mark for both fractions over a common denominator (e.g. 52/12 and 21/12, or 4/12 and 9/12); 1 mark for 2 7/12. 31/12 alone scores 1 mark (not a mixed number).

(b) Work out 6 × 2 2/3 **[1]**

> **Answer:** 16
> **Notes / working:** 6 × 2 = 12 and 6 × 2/3 = 4, so 6 × 2 2/3 = 12 + 4 = **16**. (Or 6 × 8/3 = 48/3 = 16.)

(c) Work out 5 ÷ 2/3 **[1]**

> **Answer:** 7.5 or 15/2
> **Notes / working:** Dividing by 2/3 is the same as multiplying by 3/2: 5 × 3/2 = 15/2 = **7 1/2** (7.5). Check: 7.5 × 2/3 = 5 ✓. The answer is bigger than 5 because we divide by a number less than 1.

(d) Which of these fractions is equal to a **recurring** decimal? **[1]**

- ☐ A) 3/8
- ☐ B) 7/20
- ☐ C) 5/6
- ☐ D) 9/25

Answer: C) 5/6
Notes / working: 3/8 = 0.375, 7/20 = 0.35 and 9/25 = 0.36 all terminate (their denominators have only 2s and 5s as prime factors). **5/6** = 0.8333… = $0.8\dot{3}$ recurs, because 6 = 2 × 3 has a factor 3.

---

8 **[1]**

(a) Write the correct symbol, <, > or =, in the box.
−0.45 ☐ −4/9

Answer: <
Notes / working: 4/9 = 0.444…, so −4/9 = −0.444… Now −0.45 is further below zero than −0.444…, so −0.45 is smaller: **−0.45 < −4/9**. Common mistake: thinking 0.45 > 0.444 means −0.45 > −0.444.

(b) Work out 0.6 × 0.07 **[1]**

Answer: 0.042
Notes / working: 6 × 7 = 42. There are 1 + 2 = 3 decimal places in the question, so the answer has 3: **0.042**. Estimate check: a bit less than 0.1 × 0.6 = 0.06 ✓.

(c) Work out 7.2 ÷ 0.3 **[1]**

Answer: 24
Notes / working: Multiply both numbers by 10 so the divisor is whole: 7.2 ÷ 0.3 = 72 ÷ 3 = **24**.

---

9 **[2]**

(a) A youth club raises $1500 at a fundraising evening.
The money is shared between a charity, a club trip and new equipment in the ratio
charity : trip : equipment = 5 : 3 : 2
How much money is given to the club trip? Show your working.

Answer: 450 $
Notes / working: Total parts: 5 + 3 + 2 = 10. One part: $1500 ÷ 10 = $150. Trip: 3 × $150 = **$450**. (Check: charity $750 + trip $450 + equipment $300 = $1500 ✓.)
Marking: 1 mark for 1500 ÷ 10 or 150 seen; 1 mark for 450.

(b) The club's banner is 40 cm wide and 2 m long. **[1]**
Write the ratio width : length in its simplest form.

> **Answer:** 1:5
> **Notes / working:** Use the same units first: 2 m = 200 cm. 40 : 200, divide both by 40 → **1 : 5**.
> Common mistake: 40 : 2 = 20 : 1 (units not changed).

---

10 **[1]**

(a) Simplify $3a^2 + 5a - a^2 + 2a$

> **Answer:** 2a^2+7a or 7a+2a^2 or $2a^2$+7a or 7a+$2a^2$
> **Notes / working:** Collect like terms: a^2 terms: $3a^2 - a^2 = 2a^2$; a terms: 5a + 2a = 7a. Answer **$2a^2$ + 7a**. a^2 and a are NOT like terms, so they cannot be added to make $9a^3$ or $9a^2$.

(b) Expand $2x(x^2 + 5)$ **[1]**

> **Answer:** 2x^3+10x or 10x+2x^3 or $2x^3$+10x or 10x+$2x^3$
> **Notes / working:** Multiply each term inside the bracket by 2x: $2x \times x^2 = 2x^3$ and $2x \times 5 = 10x$.
> Answer **$2x^3$ + 10x**.

(c) Factorise completely $12y^2 + 18y$ **[2]**

> **Answer:** 6y(2y+3) or 6y(3+2y) or (2y+3)6y or (3+2y)6y or 6y×(2y+3) or 6y*(2y+3)
> **Notes / working:** HCF of 12 and 18 is 6; both terms contain y. So the HCF is 6y. $12y^2 \div 6y = 2y$ and $18y \div 6y = 3$. Answer **6y(2y + 3)**. Check by expanding: $12y^2 + 18y$ ✓.
> Marking: 2 marks for 6y(2y + 3); 1 mark for a correct partial factorisation such as $6(2y^2 + 3y)$, 3y(4y + 6) or 2y(6y + 9) ('completely' means take out the HCF).

(d) Which one of these is an **equation**? **[1]**

- ☐ A) 4n + 7
- ☐ B) P = 2l + 2w
- ☐ C) y > 2
- ☐ D) 3t − 5 = 16

> **Answer:** D) 3t − 5 = 16
> **Notes / working:** 4n + 7 is an expression (no = sign). P = 2l + 2w is a formula (links several variables; it works for any values). y > 2 is an inequality. **3t − 5 = 16** is an equation: it is true for one value of t only (t = 7).

11 Solve these equations. **[2]**

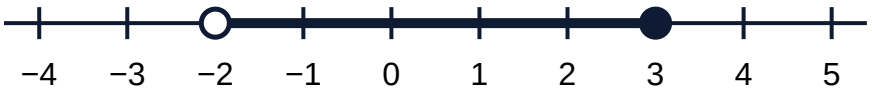
(a) 5x − 7 = 2x + 8
Show your working.

Answer: 5
Notes / working: Subtract 2x from both sides: 3x − 7 = 8. Add 7: 3x = 15. Divide by 3: **x = 5**.
Check: 5 × 5 − 7 = 18 and 2 × 5 + 8 = 18 ✓.
Marking: 1 mark for a correct first step that collects x terms or numbers (e.g. 3x − 7 = 8 or 3x = 15); 1 mark for x = 5.

(b) x/4 + 3 = 1 **[1]**

Answer: -8
Notes / working: Subtract 3: x/4 = −2. Multiply by 4: **x = −8**. Check: −8 ÷ 4 + 3 = −2 + 3 = 1 ✓.

12 The diagram shows an inequality on a number line. **[1]**



Write down the inequality shown, using x.

Answer: -2 < x <= 3 or -2 < x <=3 or -2 < x =< 3 or -2 < x =<3 or -2 < x ≤ 3 or -2 < x ≤3 or -2 < x<= 3 or -2 < x<=3 or -2 < x=< 3 or -2 < x=<3 or -2 < x≤ 3 or -2 < x≤3 or -2 <x <= 3 or -2 <x <=3 or -2 <x =< 3 or -2 <x =<3 or -2 <x ≤ 3 or -2 <x ≤3 or -2 <x<= 3 or -2 <x<=3 or -2 <x=< 3 or -2 <x=<3 or -2 <x≤ 3 or -2 <x≤3 or -2< x <= 3 or -2< x <=3 or -2< x =< 3 or -2< x =<3 or -2< x ≤ 3 or -2< x ≤3 or -2< x<= 3 or -2< x<=3 or -2< x=< 3 or -2< x=<3 or -2< x≤ 3 or -2< x≤3 or -2<x <= 3 or -2<x <=3 or -2<x =< 3 or -2<x =<3 or -2<x ≤ 3 or -2<x ≤3 or -2<x<= 3 or -2<x<=3 or -2<x=< 3 or -2<x=<3 or -2<x≤ 3 or -2<x≤3 or 3 >= x > -2 or 3 >= x >-2 or 3 >= x> -2 or 3 >= x>-2 or 3 >=x > -2 or 3 >=x >-2 or 3 >=x> -2 or 3 >=x>-2 or 3 ≥ x > -2 or 3 ≥ x >-2 or 3 ≥ x> -2 or 3 ≥ x>-2 or 3 ≥x > -2 or 3 ≥x >-2 or 3 ≥x> -2 or 3 ≥x>-2 or 3>= x > -2 or 3>= x >-2 or 3>= x> -2 or 3>= x>-2 or 3>=x > -2 or 3>=x >-2 or 3>=x> -2 or 3>=x>-2 or 3≥ x > -2 or 3≥ x >-2 or 3≥ x> -2 or 3≥ x>-2 or 3≥x > -2 or 3≥x >-2 or 3≥x> -2 or 3≥x>-2
Notes / working: An open (empty) circle means the number is NOT included: x > −2, written −2 < x. A filled circle means it IS included: x ≤ 3. Together: **−2 < x ≤ 3**.

13 **[1]**

(a) The first term of a sequence is 3/4. The term-to-term rule is 'add 1/2'.
Work out the 4th term. Give your answer as a mixed number.

Answer: 2.25
Notes / working: 3/4 → 5/4 → 7/4 → 9/4. The 4th term is 9/4 = **2 1/4**. (Adding 1/2 = 2/4 each time.)

(b) Here are the first four terms of another sequence. **[2]**

7, 4, 1, −2, …

Find an expression for the nth term.

> **Answer:** 10-3n or -3n+10 or 10-3×n or -3×n+10
> **Notes / working:** The terms go down by 3 each time, so the nth term starts with −3n. When n = 1, −3n = −3, and we need 7, so add 10. nth term = **10 − 3n** (or −3n + 10). Check n = 4: 10 − 12 = −2 ✓.
> Marking: 2 marks for 10 − 3n oe; 1 mark for −3n + c with c ≠ 10, or for 10 − 3n written as a term-to-term rule. Common mistake: n − 3 (that is the term-to-term rule idea, not the nth term).

---

14 Here is a function machine. **[1]**

input → [× 3] → [− 5] → output

(a) Find the output when the input is 4.

> **Answer:** 7
> **Notes / working:** 4 × 3 = 12, then 12 − 5 = **7**.

(b) Find the input when the output is −3. **[1]**

Give your answer as a fraction.

> **Answer:** 2/3
> **Notes / working:** Work backwards with inverse operations: −3 + 5 = 2, then 2 ÷ 3 = **2/3**. Check: 2/3 × 3 = 2, 2 − 5 = −3 ✓.

---

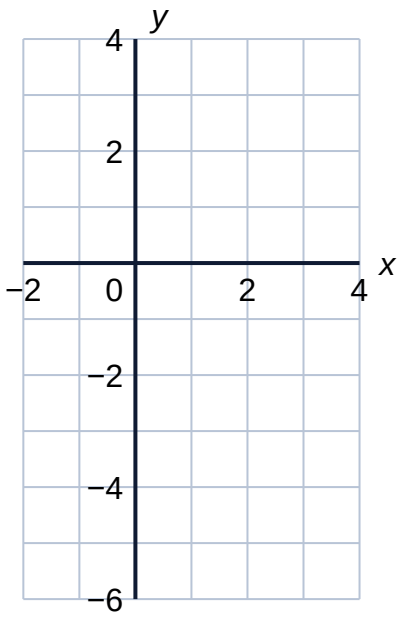
15 **[1]**

(a) Which line has gradient −2 and crosses the y-axis at (0, 5)?

- ☐ A) y = −2x + 5
- ☐ B) y = 5x − 2
- ☐ C) y = 2x + 5
- ☐ D) y = −5x + 2

> **Answer:** A) y = −2x + 5
> **Notes / working:** In y = mx + c, m is the gradient and c is the y-intercept. m = −2 and c = 5 give **y = −2x + 5**.

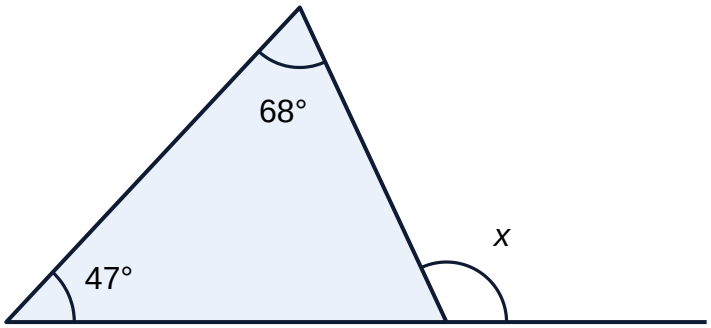
[2]



(b) Complete a table of values for y = 2x − 3 for x = −1, 0, 1, 2, 3.
Then draw the graph of y = 2x − 3 for values of x from −1 to 3 on a copy of this grid.
Draw the graph (on paper), then mark it with the mark scheme.

Answer: 1 mark for at least 3 correct points plotted from the table: (−1, −5), (0, −3), (1, −1), (2, 1), (3, 3). 1 mark for a correct straight ruled line from x = −1 to x = 3 through all the points. Do not accept a line drawn freehand that misses points.
Notes / working: Table: x = −1, 0, 1, 2, 3 gives y = −5, −3, −1, 1, 3.
Plot the five points and join them with ONE straight ruled line. Check: the line crosses the y-axis at −3 (c = −3) and goes up 2 for every 1 across (gradient 2).

16 **[1]**

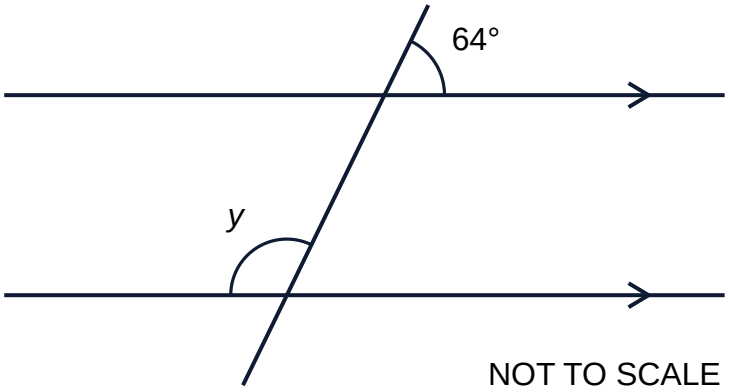


NOT TO SCALE

Work out the size of angle x.

Answer: 115 °
Notes / working: The exterior angle of a triangle equals the sum of the two interior opposite angles: x = 47° + 68° = **115°**. (Or: third angle = 180 − 47 − 68 = 65°, then x = 180 − 65 = 115°.)

17 The two lines with arrows are parallel. **[1]**



NOT TO SCALE

(a) Work out the size of angle y.

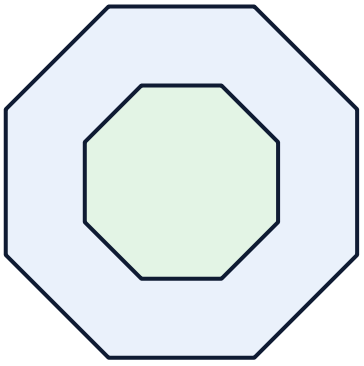
Answer: 116 °
Notes / working: The angle above the lower line, to the right of the transversal, corresponds to the 64° angle, so it is also 64°. y is on a straight line with it: y = 180° − 64° = **116°**. (Or: y and 64° are co-interior, so they add up to 180°.)

(b) Give reasons for your answer to part (a). **[1]**

> **Answer:** 1 mark for a complete reason using correct vocabulary, e.g. 'corresponding angles are equal AND angles on a straight line add up to 180°', OR 'vertically opposite angles are equal AND co-interior angles add up to 180°', OR 'alternate angles are equal and angles on a straight line add up to 180°'. Do not accept 'F angles', 'Z angles' or 'parallel lines' on its own.
> **Notes / working:** Corresponding angles are equal, so the angle at the lower crossing in the same position as 64° is 64°. Angles on a straight line add up to 180°, so y = 180° − 64° = 116°. Always use the proper words: corresponding, alternate, vertically opposite, angles on a straight line.

---

18 **[1]**



Plan view of the fountain

(a) A mosque courtyard has a fountain in its centre. From above, the fountain is a regular octagon. Write down the number of lines of symmetry and the order of rotational symmetry of a regular octagon.

> **Answer:** 1 mark for BOTH: 8 lines of symmetry AND rotational symmetry of order 8.
> **Notes / working:** A regular polygon with n sides has n lines of symmetry and rotational symmetry of order n. An octagon has 8 sides, so it has **8 lines of symmetry** (4 through opposite corners and 4 through the midpoints of opposite sides) and **rotational symmetry of order 8** (it looks the same after every turn of 360° ÷ 8 = 45°).

(b) The courtyard floor is decorated with quadrilaterals. **[1]**
Which statement is **always** true?

- ☐ A) A rhombus is a square.
- ☐ B) A trapezium is a parallelogram.
- ☐ C) A kite is a rhombus.
- ☐ D) A square is a rectangle.

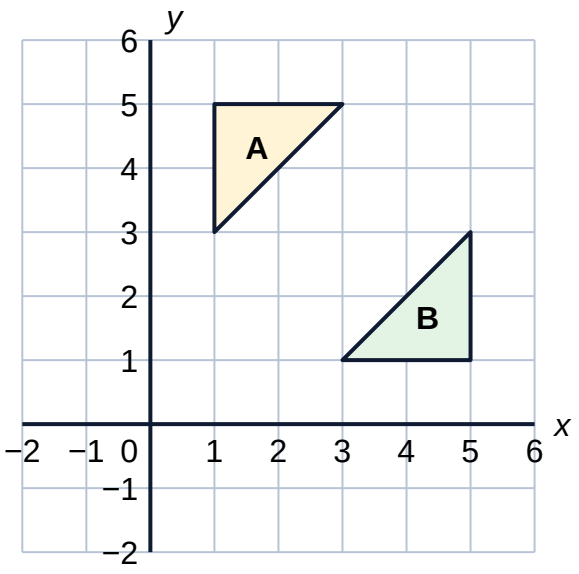
> **Answer:** D) A square is a rectangle.
> **Notes / working:** A square has four right angles and two pairs of parallel sides, so it has every property of a rectangle: **a square is always a rectangle**. A rhombus need not have right angles; a trapezium may have only one pair of parallel sides; a kite need not have four equal sides.

(c) A lantern in the courtyard is a polyhedron with 12 vertices and 18 edges. **[1]**
Use Euler's formula to work out the number of faces of the lantern.

Answer: 8
Notes / working: Euler's formula: F + V − E = 2. So F + 12 − 18 = 2 → F − 6 = 2 → **F = 8**. (This is a hexagonal prism: 2 hexagons + 6 rectangles.)

---

19 **[2]**



Describe fully the **single** transformation that maps triangle A onto triangle B.

Answer: 1 mark for 'reflection'; 1 mark for 'in the line y = x'. Also accept, for full marks, 'rotation of 180° about (3, 3)' — because this triangle is isosceles, this rotation also maps A exactly onto B. Do not accept 'flip' or 'mirror'. A combination of two transformations scores 0.
Notes / working: Reflection in the line y = x. Check each vertex: (1, 3) → (3, 1), (1, 5) → (5, 1), (3, 5) → (5, 3). Reflecting in y = x swaps the coordinates: (a, b) → (b, a). Each point and its image are the same distance from the line y = x.

---

20 In a board game, a player spins a fair spinner numbered 1, 2, 3, 4 and rolls a fair six-sided dice numbered 1 to 6. **[1]**

(a) In this game, the probability that a player lands on a 'bonus' square on a turn is 0.18.
Work out the probability that the player does **not** land on a bonus square.

Answer: 0.82
Notes / working: Landing on a bonus square and not landing on one are complementary events, so their probabilities add up to 1: 1 − 0.18 = **0.82**.

(b) The player adds the number on the spinner and the number on the dice. **[2]**
Work out the probability that the total is 7. Show your working.

Answer: 1/6
Notes / working: There are 4 × 6 = 24 equally likely outcomes (use a sample-space table: spinner 1–4 down the side, dice 1–6 across the top).
Totals of 7: (1, 6), (2, 5), (3, 4), (4, 3) → 4 outcomes.
P(total 7) = 4/24 = **1/6**.
Marking: 1 mark for 24 outcomes or the 4 correct pairs seen; 1 mark for 4/24 oe. Common mistake: counting only the possible totals 2 to 10 (9 totals) as if they were equally likely.

Paper 2 — 60 minutes — 50 marks

1 An online shop is having a sale. **[2]**

(a) A jacket normally costs $64. In the sale its price is reduced by 15%.
Work out the sale price of the jacket, in dollars. Show your working.

Answer: 54.4
Notes / working: Multiplier method: 100% − 15% = 85% = 0.85. 64 × 0.85 = **$54.40**.
Or: 15% of 64 = 9.60, then 64 − 9.60 = 54.40.
Marking: 1 mark for 64 × 0.85 or 9.6 seen; 1 mark for 54.40 (accept 54.4). Common mistake: 64 − 15 = 49.

(b) After the sale, the price of a pair of headphones goes up from $48 to $54. **[2]**
Work out the percentage increase. Show your working.

Answer: 12.5 %
Notes / working: Absolute change = 54 − 48 = $6. Percentage increase = change ÷ ORIGINAL × 100 = 6 ÷ 48 × 100 = **12.5%**.
Marking: 1 mark for 6/48 seen (or 54/48 = 1.125); 1 mark for 12.5. Common mistake: dividing by the new price, 6 ÷ 54 × 100 = 11.1%.

(c) The price of a pencil case rises from $20 to $25. **[2]**
The price of a school bag rises from $80 to $92.
Hana says, 'The price of the school bag increased more.'
Explain why Hana is correct in one way but not correct in another way. Show your working.

Answer: 1 mark for the absolute changes: bag $12 and pencil case $5, so the bag increased more in dollars. 1 mark for the percentage changes: pencil case 25% and bag 15%, so the pencil case increased more in percentage terms. Accept 25/20 = 1.25 and 92/80 = 1.15 for the second mark.
Notes / working: Absolute change: bag 92 − 80 = $12; pencil case 25 − 20 = $5. In dollars, Hana is correct: the bag went up more.
Percentage change: pencil case 5 ÷ 20 × 100 = 25%; bag 12 ÷ 80 × 100 = 15%. Compared with its original price, the pencil case went up more, so Hana is not correct in percentage terms.

2 **[2]**

(a) The online shop sells rice in two pack sizes.
Pack A: 750 g for $4.20
Pack B: 1.2 kg for $6.60
Which pack is better value for money? Show your working.

Answer: 1 mark for two values that can be compared, using the same units, e.g. $5.60 per kg and $5.50 per kg; OR 56 cents and 55 cents per 100 g; OR 178.6 g and 181.8 g per dollar. 1 mark for 'Pack B' with correct supporting values. 'Pack B' with no working scores 0.
Notes / working: Change to the same units: 750 g = 0.75 kg.
Pack A: 4.20 ÷ 0.75 = $5.60 per kg. Pack B: 6.60 ÷ 1.2 = $5.50 per kg.
Pack B costs less per kilogram, so **Pack B** is better value.
Common mistake: comparing 4.20 ÷ 750 with 6.60 ÷ 1.2 (different units).

(b) 5 m of garden hose costs $13.75. The cost is directly proportional to the length. **[1]**
Work out the cost of 8 m of the same hose.

Answer: 22 $
Notes / working: Cost of 1 m: 13.75 ÷ 5 = $2.75. Cost of 8 m: 2.75 × 8 = **$22** ($22.00). (Or 13.75 × 8/5 = 22.)

3 A passenger train runs between several towns. **[1]**

(a) In one week the train travelled 6847.5 km.
Round 6847.5 to 2 significant figures.

Answer: 6800
Notes / working: The first two significant figures are 6 and 8 (6800…). The next digit is 4, which is less than 5, so round down: **6800** km. Keep the zeros to hold the place value: 68 is wrong.

(b) The distance between two of the towns is 45 miles. **[1]**
Use 1 mile ≈ 1.6 km to change 45 miles into kilometres.

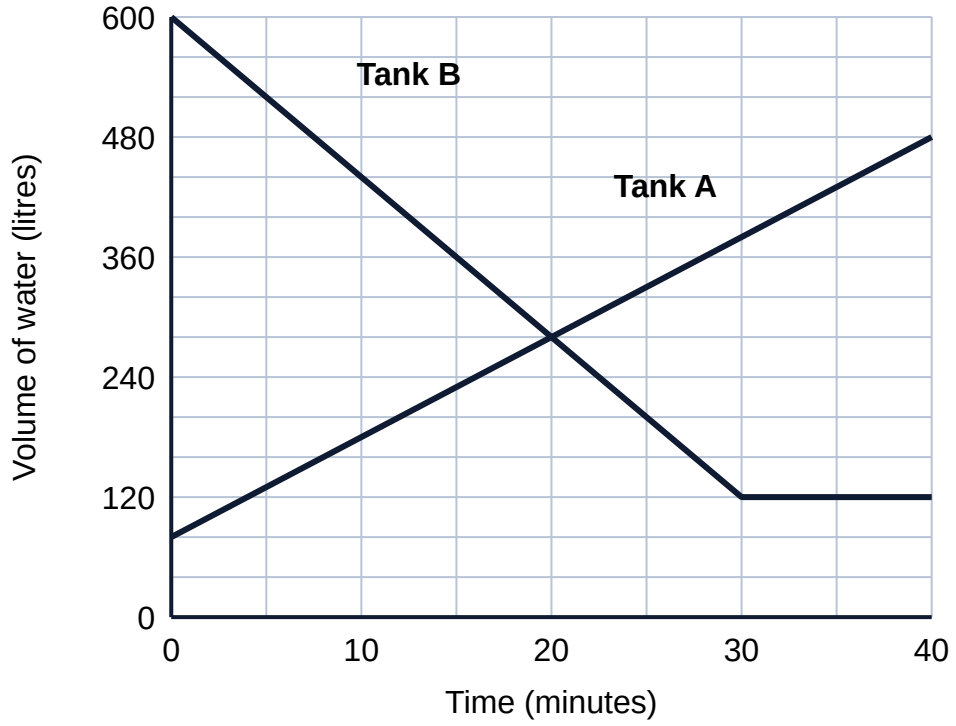
Answer: 72 km
Notes / working: 45 × 1.6 = **72 km**. A kilometre is shorter than a mile, so the number of kilometres must be bigger.

(c) The whole route is 120 km long. **[1]**
Use 1 km ≈ 5/8 mile to change 120 km into miles.

Answer: 75 miles
Notes / working: 120 × 5/8 = 600 ÷ 8 = **75 miles**. (Check with the other fact: 75 × 1.6 = 120 ✓.)

4 Two water tanks, A and B, are connected to pipes. **[1]**

The graph shows the volume of water in each tank over 40 minutes.



(a) How many litres of water are in Tank B at the start?

Answer: 600 litres
Notes / working: At time 0 the Tank B line starts at **600 litres**.

(b) After how many minutes do the two tanks hold the same volume of water? **[1]**

Answer: 20 minutes
Notes / working: The lines cross at **20 minutes** (both tanks then hold 280 litres).

(c) Describe what happens to the water in Tank B between 30 minutes and 40 minutes. Give a reason for the shape of the graph. **[1]**

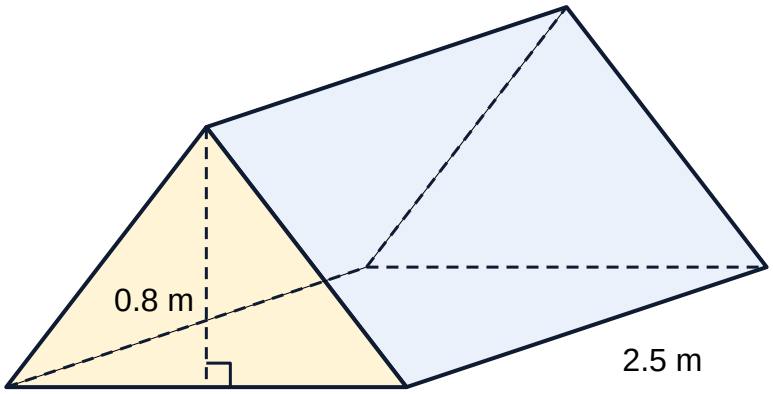
Answer: 1 mark for: the volume stays the same (at 120 litres) because water is no longer flowing out (the pipe/tap is closed), with reference to the horizontal/flat line. Do not accept 'the tank is empty'.
Notes / working: The graph is horizontal (flat) from 30 to 40 minutes, so the volume stays at **120 litres**: water has stopped flowing out of Tank B (the outlet pipe has been closed). A flat line means no change.

(d) Tank A fills at a steady rate. Use the graph to find this rate, in litres per minute, and explain what the steady rate tells you about the shape of the graph for Tank A. **[1]**

Answer: 1 mark for 10 litres per minute (e.g. (480 − 80) ÷ 40) with a statement that a steady rate gives a straight line (constant gradient). Accept 100 litres every 10 minutes.
Notes / working: Tank A goes from 80 litres to 480 litres in 40 minutes: 400 ÷ 40 = **10 litres per minute**. Because the rate is constant, the graph for Tank A is a straight line (its gradient stays the same).

---

5 **[2]**



NOT TO SCALE

(a) A water trough on a farm is a triangular prism.
Work out the volume of the trough. Show your working.

Answer: 1.2 m^3
Notes / working: Area of the triangular cross-section = ½ × base × height = ½ × 1.2 × 0.8 = 0.48 m^2.
Volume = area of cross-section × length = 0.48 × 2.5 = **1.2 m^3**.
Marking: 1 mark for 0.48 seen (or ½ × 1.2 × 0.8 × 2.5 seen); 1 mark for 1.2. Common mistake: 1.2 × 0.8 × 2.5 = 2.4 (forgetting the ½).

(b) A water trough on a farm is a triangular prism. **[1]**
1 m^3 = 1000 litres.
How many litres of water does the full trough hold?

Answer: 1200 litres
Notes / working: 1.2 × 1000 = **1200 litres**. (Follow through: 1000 × your answer to part (a).)

(c) A closed water tank is a cuboid 2 m long, 1.5 m wide and 1.2 m high. **[2]**
The outside of the tank is painted. Work out its total surface area. Show your working.

Answer: 14.4 m^2
Notes / working: There are three pairs of equal faces:
$2 \times 1.5 = 3\ m^2$; $2 \times 1.2 = 2.4\ m^2$; $1.5 \times 1.2 = 1.8\ m^2$.
Total = $2 \times (3 + 2.4 + 1.8) = 2 \times 7.2 =$ **14.4 m^2**.
Marking: 1 mark for at least two of 3, 2.4, 1.8 seen; 1 mark for 14.4. Common mistake: 7.2 (only one face of each pair).

---

6 A new courtyard is being designed for a mosque. **[2]**

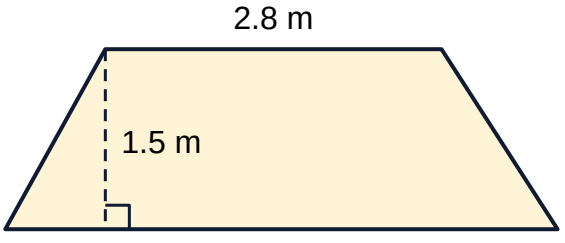
(a) A circular pool in the courtyard has a diameter of 3.5 m.
Work out the circumference of the pool. Give your answer correct to 1 decimal place.

Answer: 11 m
Notes / working: $C = \pi d = \pi \times 3.5 = 10.995\ldots$ m = **11.0 m** (1 d.p.).
Marking: 1 mark for $\pi \times 3.5$ seen or 10.99…; 1 mark for 11.0 (accept 11). Common mistake: using the formula for area, or $\pi \times 3.5 \times 2 = 22.0$ (3.5 is the diameter, not the radius).

(b) A circular path around the courtyard has a circumference of 40 m. **[1]**
Work out the diameter of the path. Give your answer correct to 1 decimal place.

Answer: 12.7 m
Notes / working: $C = \pi d$, so $d = C \div \pi = 40 \div \pi = 12.732\ldots =$ **12.7 m** (1 d.p.). π is the ratio circumference ÷ diameter.

[2]



NOT TO SCALE

(c) A marble panel on the courtyard wall is a trapezium.
Work out the area of the panel. Show your working.

Answer: 5.55 m^2
Notes / working: Area of a trapezium = ½ × (a + b) × h = ½ × (4.6 + 2.8) × 1.5 = ½ × 7.4 × 1.5 = 3.7 × 1.5 = **5.55 m^2**.
Marking: 1 mark for ½ × (4.6 + 2.8) × 1.5 seen; 1 mark for 5.55.

7 The youth club held a car wash to raise money. The stem-and-leaf diagram shows the amounts, in dollars, collected by 15 boys. **[1]**

Stem	Leaf
1	2 5 8
2	0 3 3 7 9
3	1 4 4 6
4	0 2 8

Key: stem 2, leaf 3 means $23

(a) Find the median amount.

Answer: 29 $
Notes / working: There are 15 values, so the median is the (15 + 1) ÷ 2 = 8th value. Counting in order: 12, 15, 18, 20, 23, 23, 27, **29** … Median = **$29**.

(b) Find the range of the amounts. **[1]**

Answer: 36 $
Notes / working: Range = largest − smallest = 48 − 12 = **$36**.

(c) For 15 girls at the same car wash, the median amount was $33 and the range was $18. **[2]**
Make two comparisons between the amounts collected by the boys and the girls.

Answer: 1 mark for comparing averages in context: the girls' median ($33) is higher than the boys' ($29), so on average the girls collected more. 1 mark for comparing spread in context: the girls' range ($18) is smaller than the boys' ($36), so the girls' amounts were more consistent / less spread out. FT their (a) and (b). Do not accept statements that only quote the numbers without saying what they mean.
Notes / working: Average: the girls' median, $33, is greater than the boys' median, $29, so **on average the girls collected more money**.
Spread: the girls' range, $18, is smaller than the boys' range, $36, so **the girls' amounts were more consistent** (less spread out).

8 **[1]**

(a) Altogether, the youth club raised $2400 this year. It gave $1080 of this to charity.
The club draws a pie chart to show how the $2400 was used.
Work out the angle of the sector for charity.

Answer: 162 °
Notes / working: Fraction for charity = 1080 ÷ 2400 = 0.45. Angle = 0.45 × 360° = **162°**.

(b) The club has 240 members. The leader wants to know which activities the members would like next term. He decides to ask a sample of members. **[1]**
Which sample is **least** likely to be biased?

- ☐ A) Choose 30 members at random from the full list of 240 members
- ☐ B) Ask the first 20 members who arrive on Friday evening
- ☐ C) Ask 20 members of the club's football team
- ☐ D) Ask the leader's own friends in the club

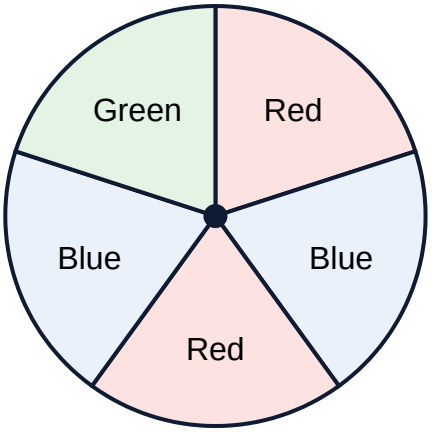
Answer: A) Choose 30 members at random from the full list of 240 members
Notes / working: A **random sample from the full list** gives every member the same chance of being chosen. The other samples only include certain kinds of member (early arrivals, footballers, the leader's friends), whose views may not represent the whole club.

(c) The club has 240 members. The leader wants to know which activities the members would like next term. He decides to ask a sample of members. **[1]**
Give one advantage and one disadvantage of asking a sample instead of asking all 240 members.

Answer: 1 mark for BOTH: an advantage (e.g. quicker / cheaper / easier to collect and analyse) AND a disadvantage (e.g. may not represent all members / results less reliable / some opinions are missed).
Notes / working: Advantage: a sample is **quicker and easier** to collect and analyse than asking everyone.
Disadvantage: the sample **might not represent all 240 members**, so the results may be less reliable than asking everybody.

9 Karim makes a spinner for a board game. It has five equal sections: 2 red, 2 blue and 1 green. **[1]**
He spins it 200 times. The results are:

Colour	Red	Blue	Green
Frequency	74	85	41



(a) Work out the experimental probability of the spinner landing on red.

Answer: 0.37 or 74/200 or 37/100
Notes / working: Experimental probability = frequency ÷ number of trials = 74 ÷ 200 = **0.37** (= 37/100).

(b) Write down the theoretical probability of the spinner landing on red. **[1]**

Answer: 0.4 or 2/5
Notes / working: 2 of the 5 equal sections are red, so P(red) = **2/5** = 0.4.

(c) Karim says, 'My spinner must be biased, because 0.37 is not the same as my answer to part (b).' **[1]**
Is Karim correct? Explain your answer.

Answer: 1 mark for 'No' (or 'not necessarily') with a reason, e.g. 0.37 is close to 0.4; experimental results vary and are not expected to match exactly; he should do more trials to check. Do not accept 'Yes'.
Notes / working: Not necessarily. Experimental probabilities change from one experiment to another, and 0.37 is close to 0.4 (blue, 0.425, and green, 0.205, are also close to 0.4 and 0.2). To test for bias he should spin it many more times: with a large number of trials, the experimental probability should get closer to the theoretical one if the spinner is fair.

10 **[1]**

(a) At the youth club quiz night, an adult ticket costs $x and a child ticket costs $4 less than an adult ticket.
A family buys 2 adult tickets and 3 child tickets. They pay $43 altogether.
Show that 5x − 12 = 43.

Answer: 1 mark for a correct expression for the total cost, e.g. 2x + 3(x − 4) = 43, expanded to reach 5x − 12 = 43 with no errors.
Notes / working: A child ticket costs x − 4 dollars. Total: 2x + 3(x − 4) = 43 → 2x + 3x − 12 = 43 → **5x − 12 = 43**.

(b) At the youth club quiz night, an adult ticket costs $x and a child ticket costs $4 less than an adult ticket. **[1]**
A family buys 2 adult tickets and 3 child tickets. They pay $43 altogether.
Solve the equation to find the cost of an adult ticket.

Answer: 11 $
Notes / working: 5x − 12 = 43 → 5x = 55 → **x = 11**. An adult ticket costs $11 (and a child ticket $7). Check: 2 × 11 + 3 × 7 = 22 + 21 = 43 ✓.

(c) The online shop uses this formula for the delivery cost, C dollars, of a parcel with a mass of w kilograms: **[1]**
C = 4.5 + 0.8w
Work out the delivery cost for a parcel with a mass of 12.5 kg.

Answer: 14.5 $
Notes / working: C = 4.5 + 0.8 × 12.5 = 4.5 + 10 = **$14.50**. Multiply before adding.

(d) The online shop uses this formula for the delivery cost, C dollars, of a parcel with a mass of w kilograms: **[1]**
C = 4.5 + 0.8w
Rearrange the formula to make w the subject.

Answer: 1 mark for w = (C − 4.5) ÷ 0.8 oe, e.g. w = (C − 4.5)/0.8, w = 1.25(C − 4.5), w = 1.25C − 5.625. Brackets must be correct: do not accept w = C − 4.5 ÷ 0.8.
Notes / working: Subtract 4.5 from both sides: C − 4.5 = 0.8w. Divide both sides by 0.8: **w = (C − 4.5) ÷ 0.8** (which is the same as 1.25(C − 4.5)). Check with part (c): (14.5 − 4.5) ÷ 0.8 = 12.5 ✓.

(e) The youth club has p members. Two-thirds of the members go on a trip, and 5 leaders go with them. **[1]**

Write an expression, in terms of p, for the total number of people who go on the trip.

> **Answer:** 2/3p+5 or (2/3)p+5 or 2p/3+5 or (2p)/3+5 or 5+2/3p or 5+(2/3)p or 5+2p/3 or 2/3×p+5 or ⅔p+5 or 2/3*p+5
> **Notes / working:** Two-thirds of p is (2/3)p, also written 2p/3. Add the 5 leaders: **(2/3)p + 5**. Common mistake: 2/3 + p + 5 or 2/3(p + 5).

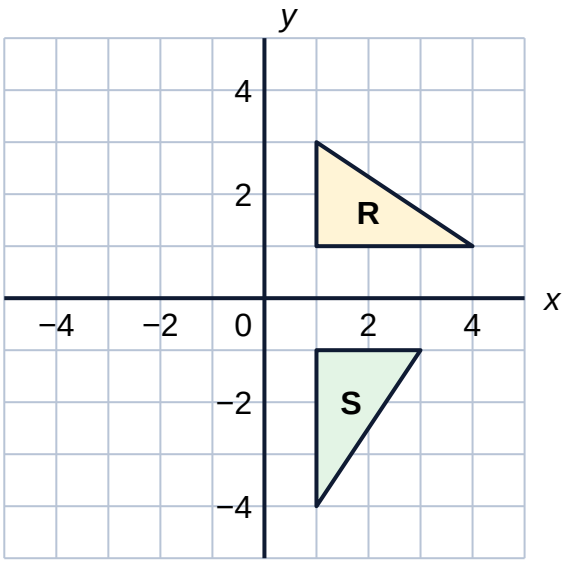
---

11 **[1]**

(a) Triangle T has vertices at (2, 3), (3, 3) and (2, 5).
T is enlarged by scale factor 3, with centre of enlargement (1, 1).
Write down the coordinates of the image of the vertex (2, 5).

> **Answer:** (4,13) or 4,13
> **Notes / working:** From the centre (1, 1) to (2, 5) is 1 right and 4 up. Multiply by the scale factor 3: 3 right and 12 up. From (1, 1): (1 + 3, 1 + 12) = **(4, 13)**. Common mistake: multiplying the coordinates by 3 from the origin, giving (6, 15).

[2]



(b) Describe fully the single transformation that maps triangle R onto triangle S.

> **Answer:** 1 mark for 'rotation' with '90° clockwise' (accept 270° anticlockwise); 1 mark for 'centre (0, 0)' or 'about the origin'. Only 1 mark in total if two different transformations are given.
> **Notes / working: Rotation, 90° clockwise, centre (0, 0).** Check: (1, 1) → (1, −1), (4, 1) → (1, −4), (1, 3) → (3, −1). A rotation needs three facts: the centre, the angle and the direction. Use tracing paper to check: trace R, put your pencil on (0, 0) and turn a quarter-turn clockwise.

12 **[1]**

(a) On a plan of the mosque courtyard, the bearing of the minaret from the gate is 125°.
Work out the bearing of the gate from the minaret.

Answer: 305 °
Notes / working: The back bearing differs by 180°: 125° + 180° = **305°**. (If the bearing is more than 180°, subtract 180° instead.) Draw a sketch with north lines at both points to check.

(b) On the same plan, a straight path goes from (−3, 5) to (7, −1). A lamp is placed at the midpoint of the path. **[1]**
Find the coordinates of the lamp.

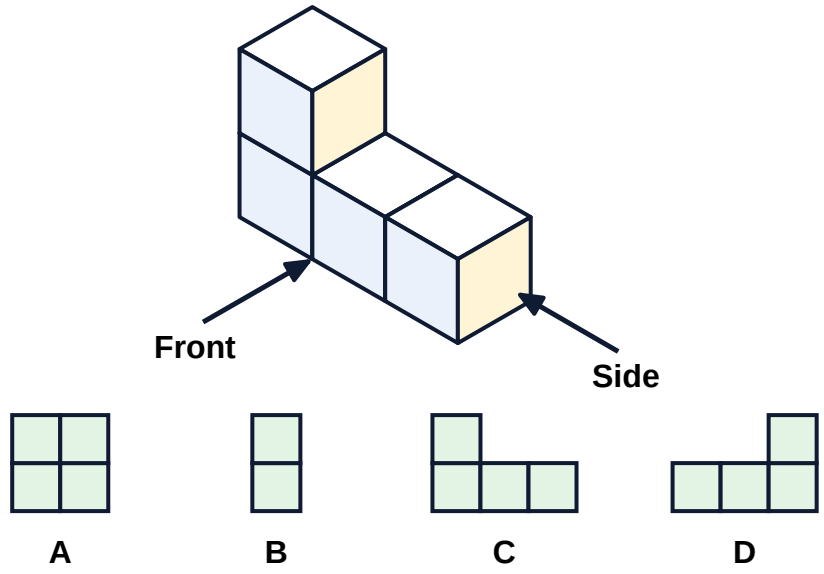
Answer: (2,2) or 2,2
Notes / working: Midpoint = (mean of the x-coordinates, mean of the y-coordinates) = ((−3 + 7) ÷ 2, (5 + (−1)) ÷ 2) = (4 ÷ 2, 4 ÷ 2) = **(2, 2)**.

13 A triangular flower bed in the courtyard is drawn to scale as triangle ABC with AB = 7 cm, AC = 5 cm and BC = 6 cm. **[2]**

On paper, use a ruler and compasses only to construct triangle ABC. Leave in your construction arcs.

Answer: 1 mark for AB drawn 7 cm (±1 mm) with an arc of radius 5 cm from A or 6 cm from B. 1 mark for a complete correct triangle with both arcs shown and sides within ±1 mm. Do not award the second mark if no arcs are visible.
Notes / working: 1. Draw AB = 7 cm with a ruler.
2. Open the compasses to 5 cm; with the point on A, draw an arc above AB.
3. Open the compasses to 6 cm; with the point on B, draw an arc that crosses the first arc. The crossing point is C.
4. Join AC and BC with a ruler. Do not rub out the arcs: they show your method.
Check: measure AC = 5 cm and BC = 6 cm (angle A should be about 57°).

14 The diagram shows a shape made from four cubes, and four drawings A, B, C and D. **[1]**



(a) Which drawing is the **front elevation** of the shape (the view in the direction of the arrow marked Front)?

- ☐ A) A
- ☐ B) B
- ☐ C) C
- ☐ D) D

Answer: C) C
Notes / working: From the front you see the long side: three squares along the bottom and one square on top of the LEFT-hand cube, because the tall stack is at the far left of the front face. That is **drawing C**. D is the same shape reversed (as if seen from the back); B is the side elevation; A is not a view of this shape.

(b) On squared paper, draw the plan (the view from above) of the shape. **[1]**

Answer: 1 mark for a row of three squares (1 by 3 rectangle divided into three equal squares). Accept with or without the internal lines. Do not accept an L shape.
Notes / working: From above you see the tops of the three cubes in the bottom row; the top cube sits directly above the left-hand one, so it does not change the outline. The plan is **a row of three squares**.